an einem seltsamen Ort

Final Project Documentation

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Project Link: http://leahiscoding.github.io/CART263/finalProject
GitHub Repository: https://github.com/leahiscoding/CART263/tree/main/finalProject

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1. Project Overview

“an einem seltsamen Ort,” which translates to“a Strange Place," is an interactive website inspired by German techno music, night culture, and the feeling of entering an unfamiliar digital environment.

The idea came to mind when I was on a run while listening to German techno music, and I realized that I wanted to create something in response to the music. I wanted the website to feel strange, playful, and fun.

The website includes different interactive components, such as Face-Mesh, Handpose, and Speech Recognition. These sections let the user interact with the website through the face, hand, and voice, turning the user’s body into the controller.

This project explores my interpretation of German techno and nightlife culture. Growing up in East Asia and Canada, this scene is not something I am familiar with. However, after visiting Berlin a couple of years ago and listening to more techno music recently, and especially since starting Computation Arts, I have become more interested in electronic music. I wanted to create a website that could go along with the music and the nightlife culture as I imagined it.

2. Project Intentions

My main intention was to create a website that feels like entering a strange and unfamiliar place. I wanted users to feel curious, slightly confused, and playful while exploring the project.

I was particularly interested in making a website that feels more like an environment than a conventional webpage. Many websites are designed to be clear and useful, but I wanted this project to focus more on atmosphere, sound, and interactive experience instead.

I also wanted to include artificial intelligence and machine learning in a way that connected to what I learned. Rather than using AI only to generate content, I wanted to use machine-learning models as part of the interaction. This led me to explore features such as face tracking, hand tracking, and speech recognition.

The goal of the project is simple: to let the user have fun, listen to music, and explore different ways of interacting with the website. The project is still relatively simple, but it became a foundation for something I could expand more in the future.

3. User Experience

When users first enter the website, they see the title “an einem seltsamen Ort” and are prompted to turn on audio for a better experience. This is essential because the project started from music, and the sound helps create the mood.

After pressing start, the user can choose between three interactive options: Face-Mesh, Handpose, and Speech Recognition.

Face-Mesh uses the face as an input. This connects the user’s facial movement to the digital space. Meanwhile, Handpose uses the hand and gesture as input. This makes the interaction feel more physical than only clicking with a mouse. Finally, Speech Recognition uses the voice as input, connecting sound, language, and interaction.

Together, these elements encourage users to engage with the website through their body and voice, transforming it from a static webpage into a strange, immersive digital environment.

4. Development Process

At the beginning, I did not have a clear idea of what I wanted to make. The concept started while I was running and listening to German techno music. The sound and atmosphere of the music helped me imagine the mood of the website.

I then used Figma for the visual design, aiming for it to feel strange, abstract, and fun. I also used Figma MCP to help bring CSS values into my project.

Originally, I wanted to use PyTorch because I had experience with it from another class. However, after researching how PyTorch could be used with JavaScript, I realized that it was not the best option for a browser-based project. Because the website needed to run on GitHub Pages, it was not practical for my project.

I then looked into TensorFlow.js and MediaPipe. I thought they could be useful for face or hand tracking, but when I tested examples on my laptop, they loaded very slowly. They worked, but were not practical for my project.

Ultimately, I decided to use ml5.js. It was easier to integrate with p5.js and better for creative coding. I imported ml5.js code into my files and connected it with code I had used in a previous semester project. In this way, I built the project by expanding something I had already made before.

Through this process, I developed a stronger understanding of JavaScript. At first, it felt like I had many separate pieces without knowing how they fit together. As I worked through the project, I began to see how HTML, CSS, JavaScript, p5.js, and ml5.js could connect and function as a cohesive system.

5. Use of Generative AI

During development, I used generative AI tools such as Codex and Claude to support my workflow. They were helpful for tasks like importing Figma MCP values, adjusting CSS, troubleshooting errors, and understanding how different parts of the code could connect.

At the same time, I tried to write as much of the project as I could on my own. I regularly reviewed what I had learned from class and previous projects. When AI tools suggested code or solutions, I used them as support, but I still tested, edited, and reviewed the code myself.

I did not use generative AI to replace my learning. I used it more like a support system when I was stuck. The final project still reflects my own concept, design direction, technical choices, and learning process.

6. Technical Implementation, Libraries, and Tools

This project was built using HTML, CSS, JavaScript, p5.js, ml5.js, Figma, Figma MCP, Codex, Claude, and GitHub Pages.

The overall structure of the website was created with HTML, which organizes the main sections of the project including the start page, the model selection page, and the different interaction pages. It also defines basic elements such as buttons, text, and links.

The visual style was developed using CSS. It controls layout, color, typography, spacing, and overall atmosphere. I wanted the website to feel strange, playful, and connected to night culture, and CSS played an important role in shaping the mood.

JavaScript manages the interaction and behavior of the site. It allows the website to respond to user input and controls transitions between sections. For example, moving from the start page to the selection screen, and then to the Face Mesh, Handpose, or Speech Recognition experiences.

p5.js was used for creative coding and interactive visuals, providing a more expressive, design oriented way to work with graphics. ml5.js was used for the machine learning components, as it integrates well with p5.js and is accessible for creative coding. It enabled interactions based on face tracking, hand gestures, and voice input.

Figma was used during the design phase to plan the visual layout before development. Figma MCP helped translate design values into CSS, creating a smoother connection between design and implementation.

Codex and Claude were used as support tools throughout development. They assisted with troubleshooting, refining CSS, working with Figma MCP, and understanding how different parts of the code could connect. However, I tested, reviewed, and edited all the code myself.

GitHub Pages was used to publish the final project, allowing it to run directly in the browser without requiring any installation.

Two key technical features define the structure of the project. The first is the page transition system, where JavaScript controls which section is visible based on user interaction. The second is the integration of ml5.js models, which allow users to interact through face, hand, and voice input. These features connect the user's physical actions to the digital environment, reinforcing the concept of entering a strange, interactive space.

7. Challenges

One of the biggest challenges was figuring out how to connect different parts of the project. I had pieces of code from class, previous projects, online research, and AI-assisted troubleshooting, but it was difficult to make everything work together.

Another challenge was choosing the right machine learning tool. I initially looked at PyTorch, then TensorFlow.js and MediaPipe, but they weren't practical for this project. Choosing ml5.js was a better decision because it fit better with p5.js and my current skill level.

JavaScript was also challenging. Sometimes I did not understand why the code worked or why it failed. Through testing and revising, I became more comfortable with reading and changing code.

I also had to balance my original idea with what was realistic. While I wanted the website to have more complex features, I had to work with the time, performance, and skills I had. The final project is simpler than what I first imagined, but I still learned a lot from it.

8. Reflection and Conclusion

Through this project, I learned a lot about JavaScript, machine learning libraries, and interactive web design. At the beginning, I felt anxious because I didn't fully understand how everything worked.

As I continued working, I became more comfortable connecting different parts of the project. I learned how HTML, CSS, JavaScript, p5.js, and ml5.js could function together. I also learned that technical plans often evolve during the process, and that choosing a simpler tool can sometimes be the better decision.

Although the project is still simple, I am happy with it. While it began from a personal moment of listening to German techno music while running, it became an interactive website exploring music, night culture, digital strangeness, and body-based interaction.

In the future, I would like to build on this project by making deeper interactions, especially with music. I'm interested in creating stronger visual or audio responses, utilizing face, hand, and voice inputs.

Overall, this project helped me become more confident in working with JavaScript and in combining design, code, sound, and machine learning.

9. Screenshots and Documentation

This section includes screenshots that document important parts of the project. The screenshots show the user flow, visual design, interaction pages, and development process.

Figure 1. Opening Page

Figure 2. Model Selection Page

Figure 3. Face-Mesh Section

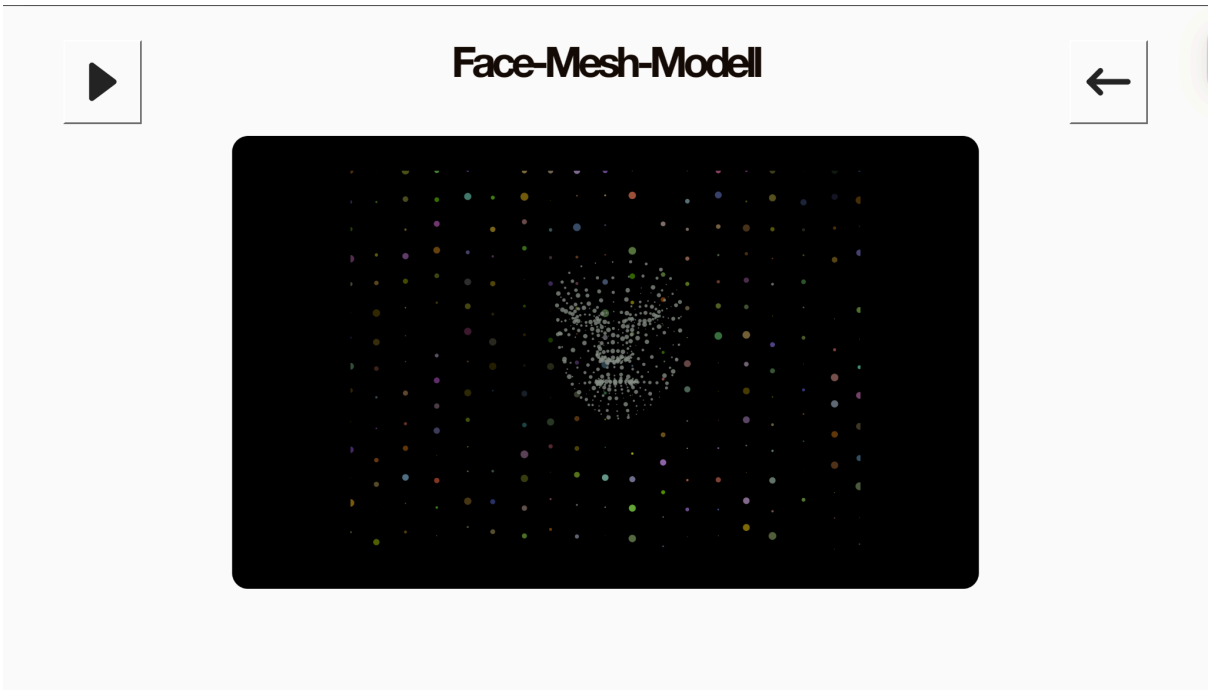


Figure 4. Handpose Section

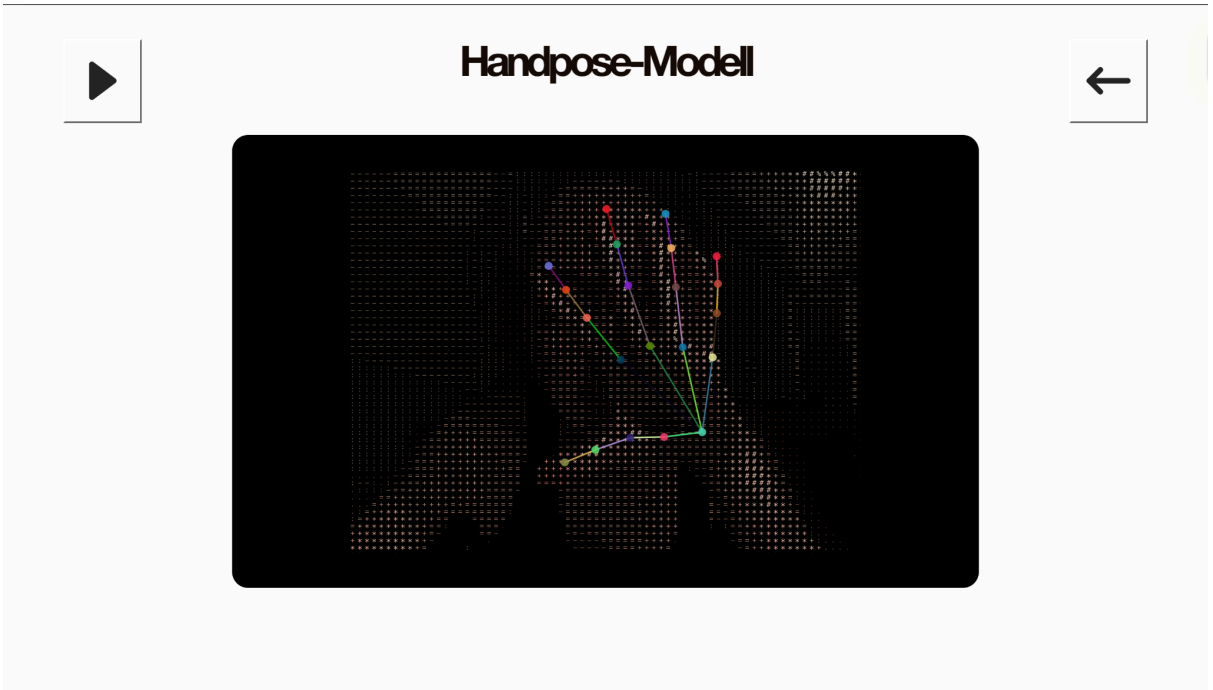


Figure 5. Speech Recognition Section

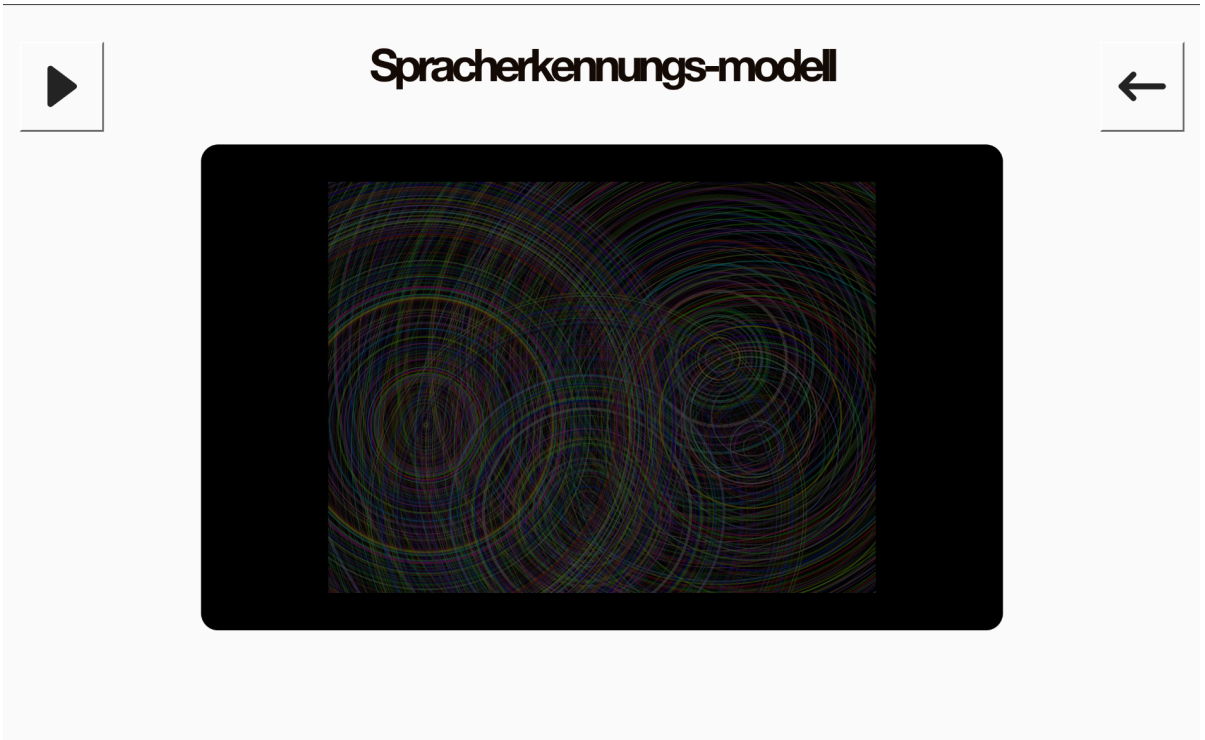
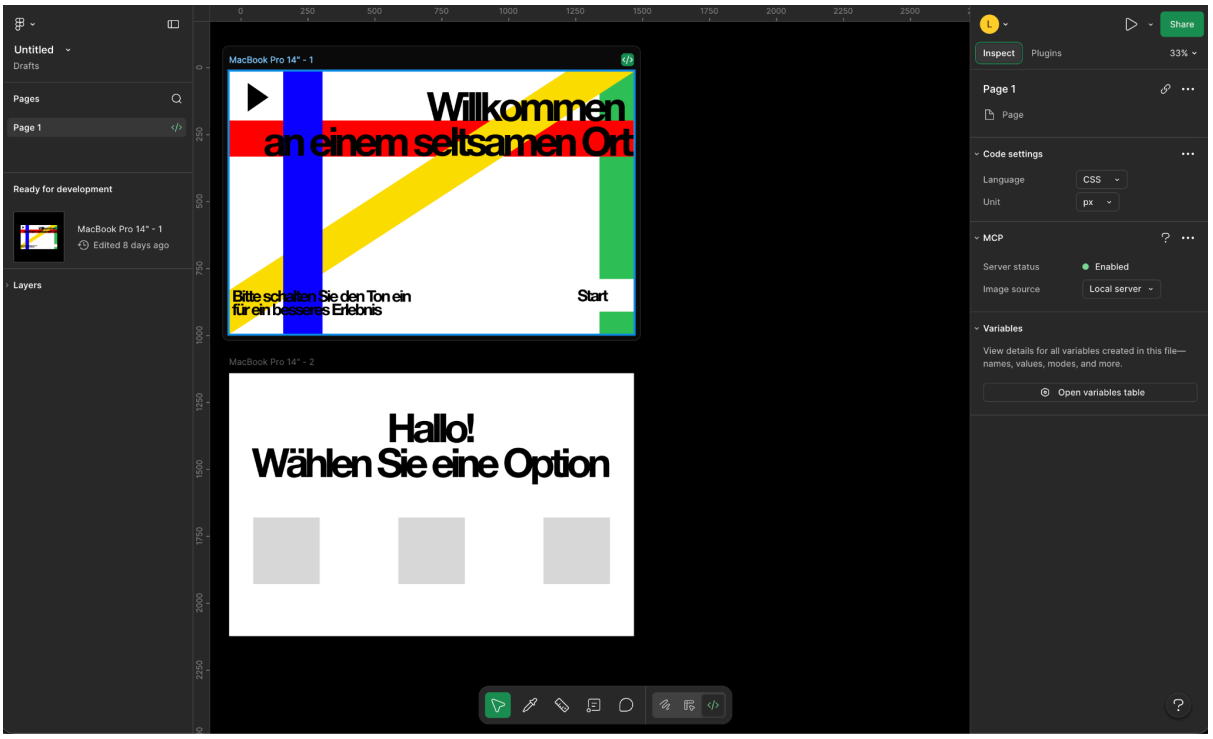


Figure 6. Figma Process Screenshot



10. References and Inspirations

Reference 1: Why Germany Loves Techno: The Culture, Clubs and History

This article by FERAL Clothing discusses Germany's relationship with techno culture, especially its connection to clubs, music history, nightlife, and underground communities. The article helped me think about techno not only as a music genre, but also as a cultural space connected to atmosphere, darkness, repetition, and collective experience.

This reference inspired my project because I wanted my website to feel like entering an unfamiliar digital nightlife space. I was especially interested in how techno culture can create a strong mood through sound, darkness, and movement. My project does not directly recreate German techno culture, but this reference helped me think about how music can shape a visual and interactive environment. It influenced the overall mood of the website and helped me connect the sound-based inspiration of the project to the feeling of entering a strange place.

Reference 2: Machine Learning Made Easy With ml5.js

This article by Reid Watson explains how ml5.js makes machine learning more accessible for creative coding and browser-based projects. It introduces ml5.js as a tool that allows artists, designers, and beginner programmers to use machine-learning models without needing to build everything from the ground up.

This reference influenced the technical direction of my project. Originally, I researched PyTorch, TensorFlow.js, and MediaPipe, but those tools felt more difficult or less practical for my browser-based project. Reading about ml5.js helped me understand that machine learning could be used in a more approachable way inside creative web projects. It inspired me to use face, hand, and voice input as part of the website experience. This helped me think about the user not only as someone clicking through a page, but as someone whose body and voice could become part of the interaction.

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